

tests/regression/ proxy_conn_verdict_test.sh

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Overview

Standing §11.4.135 regression guard for `proxy_conn_verdict()` — the client-side through-proxy connectivity classifier added to `tests/lib/evidence.sh` in **BUGFIX-0014**. It proves the classifier never commits either of the two anti-bluff sins the connectivity checks used to be capable of:

- a **§11.4.1 false-FAIL** — scoring a healthy proxy as FAIL because the *site* the probe targeted was unreachable (a third-party / local-internet outage the proxy cannot be blamed for); and
- a **§11.4.68 fail-open** — masking a *real* proxy defect as a benign SKIP.

Why this guard exists

Before BUGFIX-0014, `tests/verify-proxy.sh` and `tests/final-verify.sh` classified each through-proxy check as `code == expected -> PASS else FAIL`. That single-axis logic could not tell "the proxy is broken" apart from "the site is momentarily down", so a blip on `connectivitycheck.gstatic.com / www.google.com` — or a host with no egress — hard-FAILED a perfectly healthy proxy. Non-deterministic (§11.4.50), not re-runnable (§11.4.98).

The fix classifies with a second axis (does the site respond **directly**?) plus a port-liveness gate:

proxy in expected	direct in expected	port listening	verdict
yes	—	—	PASS
no	yes	—	FAIL (site reachable directly, proxy can't serve it — a real defect whether the port is up-but-broken OR the proxy crashed; a positive direct signal

proxy in expected	direct in expected	port listening	verdict
			out-ranks the port probe, §11.4.68 not fail-open)
no	no	yes	SKIP:network_unreachable_external (site outage — no false-FAIL)
no	no	no	SKIP:topology_unsupported (proxy absent AND no network signal to substantiate a FAIL)

Prerequisites

- POSIX sh; tests/lib/evidence.sh present (the guard **sources** it — it drives the real function, never a re-implementation).
- No network: every case is a pure-function truth-table assertion.

Usage

```
# GREEN standing guard (default): PASS iff the classifier's full
# truth table is correct.
tests/regression/proxy_conn_verdict_test.sh

# RED reproduction: PASS iff the PRE-FIX replica (code != expected
# => FAIL) FAILs an
# outage tuple – the false-FAIL demonstrated on the pre-fix logic
# (§11.4.115).
RED_MODE=1 tests/regression/proxy_conn_verdict_test.sh
```

Exit 0 = PASS, 1 = FAIL. Evidence is written under qa-results/
regression/proxy_conn_verdict/.

§11.4.115 polarity switch (RED_MODE)

- RED_MODE=0 (default) — GREEN guard: sources evidence.sh and asserts proxy_conn_verdict returns PASS / FAIL / SKIP:network_unreachable_external / SKIP:topology_unsupported across the canonical truth table (single- and multi-code expected).
- RED_MODE=1 — reproduces the defect: runs the pre-fix code != expected => FAIL replica against the outage tuple (proxy=000, expected=204) and asserts it FAILs. A RED that cannot reproduce is a §11.4.7 finding.

§1.1 paired mutation

Mutating `proxy_conn_verdict`'s outage branch in `evidence.sh` (`SKIP:network_unreachable_external` → `FAIL`, reintroducing the false-`FAIL`) makes the GREEN guard `FAIL` with a **real value-assertion mismatch** (`proxy_conn_verdict 000 000 204 yes -> FAIL (want SKIP:network_unreachable_external)`), not a parse error or panic. Restoring the file byte-identically (md5-verified) returns the guard to `PASS`. This proves the guard tracks behaviour, not syntax.

Edge cases

- **Site outage vs proxy defect** — the decisive discriminator is whether the site answers *directly*. If neither the proxy nor a direct fetch reaches it, the site is the problem → `SKIP`, not `FAIL`.
- **Crashed / absent proxy port** — a not-listening port does NOT auto-`SKIP`. If the site is reachable **directly** (proof the network works), a dead proxy is a real defect → `FAIL` (§11.4.68 not fail-open). Only when the site is *also* unreachable and nothing proves the proxy should be serving does it become `SKIP:topology_unsupported` (an unprovable-`FAIL`, honestly skipped).
- **Multi-code expectations** — expected is a space-separated allow-list (e.g. `'200 301 302'`); `_code_in` matches membership.

Related scripts

- Fix under guard: `tests/lib/evidence.sh` (`proxy_conn_verdict / _code_in / port_is_listening`).
- Consumers: `tests/verify-proxy.sh`, `tests/final-verify.sh` (`conn_check`).
- Unit truth table: `tests/lib/evidence_selftest.sh`.
- Sibling sink-side guard: `tests/regression/external_egress_verdict_test.sh` (BUGFIX-0012).
- Ledger: `docs/issues/fixed/BUGFIXES.md` — BUGFIX-0014.

Last verified

2026-07-01 — GREEN PASS, RED PASS (reproduces), §1.1 mutation FAILs + restores byte-identical, unit selftest 37/37 (incl. crashed-proxy FAIL + `_code_in` exact-match cases from the independent-review reconciliation), `verify-proxy.sh` + `final-verify.sh` each 4 PASS + 1 SKIP (exit 0) live.